AI Powered Legal Advisor and Analyzer



Actor Workflows, Actions and Dependencies

Workflow diagrams for the User, Lawyer, Admin and AI Services

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Fall 2026

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Table of Contents

1 Actors, Actions and Workflow Overview **7**

1.1 Actors . . . 7

1.2 Diagram Notation . . . 8

1.3 Workflow Summary . . . 8

1.4 End-to-End Workflow Across Actors . . . 9

1.5 Actor-Action Matrix . . . 11

2 User (Citizen) Workflows **13**

2.1 User Actions and Dependencies . . . 13

2.2 U-WF1: Account Access and Profile Management . . . 16

2.3 U-WF2: Create Case and Clarify Facts . . . 17

2.4 U-WF3: Obtain AI Legal Guidance . . . 18

2.5 U-WF4: Upload Documents and Complete Evidence Checklist . . . 19

2.6 U-WF5: Find a Lawyer and Request a Consultation . . . 21

2.7 U-WF6: Track Case, Communicate and Review . . . 22

3 Lawyer Workflows **23**

3.1 Lawyer Actions and Dependencies . . . 23

3.2 L-WF1: Registration and Verification . . . 25

3.3 L-WF2: Public Profile and Availability Management . . . 26

3.4 L-WF3: Handle Consultation Requests . . . 27

3.5 L-WF4: Work on a Shared Case . . . 28

4 Admin Workflows **29**

4.1 Admin Actions and Dependencies . . . 29

4.2 A-WF1: Lawyer Verification . . . 31

4.3 A-WF2: Legal Knowledge Base Management . . . 32

4.4 A-WF3: User and Lawyer Account Management . . . 33

4.5 A-WF4: Content Moderation . . . 34

4.6 A-WF5: Monitoring and Audit 35

5 System (AI Services) Workflows **36**

5.1 S-WF1: RAG Guidance Pipeline 37

5.2 S-WF2: Document Processing Pipeline 38

5.3 S-WF3: Knowledge Base Ingestion Pipeline 39

5.4 S-WF4: Notification Dispatch 40

6 Cross-Actor Dependencies and Status Lifecycles **41**

6.1 Dependency Graph 41

6.2 Case Status Lifecycle 44

6.3 Consultation Status Lifecycle 45

6.4 Lawyer Verification Lifecycle 46

6.5 Legal Source Lifecycle 47

List of Figures

1.1 End-to-end swimlane workflow across all actors 10

2.1 Actions available to the User 15
2.2 U-WF1 Account access and profile management 16
2.3 U-WF2 Create case and clarify facts 17
2.4 U-WF3 Obtain AI legal guidance 18
2.5 U-WF4 Upload documents and complete evidence checklist 20
2.6 U-WF5 Find a lawyer and request a consultation 21
2.7 U-WF6 Track case, communicate and review 22

3.1 Actions available to the Lawyer 24
3.2 L-WF1 Registration and verification 25
3.3 L-WF2 Public profile and availability management 26
3.4 L-WF3 Handle consultation requests 27
3.5 L-WF4 Work on a shared case 28

4.1 Actions available to the Admin 30
4.2 A-WF1 Lawyer verification 31
4.3 A-WF2 Legal knowledge base management 32
4.4 A-WF3 User and lawyer account management 33
4.5 A-WF4 Content moderation 34
4.6 A-WF5 Monitoring and audit 35

5.1 S-WF1 RAG guidance pipeline 37
5.2 S-WF2 Document processing pipeline 38
5.3 S-WF3 Knowledge base ingestion pipeline 39
5.4 S-WF4 Notification dispatch 40

6.1 Dependency graph between workflows and actors 42
6.2 Case status lifecycle 44

6.3 Consultation status lifecycle 45
6.4 Lawyer verification lifecycle 46
6.5 Legal source lifecycle 47

List of Tables

1.1 Workflows by actor 8
1.2 Actor-action matrix 11
2.1 Actions of the User and their dependencies 13
3.1 Actions of the Lawyer and their dependencies 23
4.1 Actions of the Admin and their dependencies 29
6.1 Key dependencies 43

Chapter 1 — Actors, Actions and Workflow Overview

This document presents the workflows of the *AI Legal Advisor and Analyzer* from the point of view of each actor. For every actor it lists the actions the actor can perform, the dependencies each action has on other actors or on system state, and a workflow (activity) diagram for every workflow the actor takes part in. It extends the use cases (UC-01 to UC-12) and functional requirements (UFR, LFR, AFR) of Chapter 3 of the project report. All workflows describe the planned system.

1.1 Actors

Actor	Type	Portal	Responsibility
User (Citizen)	Primary	User portal	Describes legal problems, receives grounded guidance, uploads evidence, finds lawyers, requests consultations and tracks cases.
Lawyer	Primary	Lawyer portal	Maintains a verified public profile, handles consultation requests, works on shared cases and posts updates.
Admin	Primary	Admin portal	Verifies lawyers, controls the legal knowledge base, manages accounts, moderates content and monitors activity.
AI Services (System)	Secondary	None (backend)	Embedding model, Qdrant retrieval, open-source LLM, citation checker, PyMuPDF and OCR that execute the automated steps.
Email / Notification Service	Secondary	None (backend)	Sends verification, password-reset and status notifications.

1.2 Diagram Notation

All workflow diagrams use the same notation:

- **Rounded box:** start or end of a workflow.
- **Rectangle:** an action performed by the actor or the system.
- **Hexagon:** a decision or check (labelled exits such as Yes and No).
- **Solid arrow:** the normal flow of control.
- **Dashed arrow:** a dependency on another actor's workflow or on system state.
- **Coloured lanes:** in swimlane diagrams, which actor performs each step.

Each workflow is labelled with an identifier (for example U-WF2). Each action is labelled with an identifier (for example U-A05) that is used in the action tables and the dependency analysis.

1.3 Workflow Summary

Table 1.1: Workflows by actor

Workflow ID	Workflow	Actor	Related use case	Section
U-WF1	Account access and profile management	User	UC-01	2.2
U-WF2	Create case and clarify facts	User	UC-02	2.3
U-WF3	Obtain AI legal guidance	User	UC-03	2.4
U-WF4	Upload documents and complete evidence checklist	User	UC-04	2.5
U-WF5	Find a lawyer and request a consultation	User	UC-05, UC-06	2.6
U-WF6	Track case, communicate and review	User	UC-07	2.7
L-WF1	Registration and verification	Lawyer	UC-01, UC-08	3.2
L-WF2	Public profile and availability management	Lawyer	UC-08	3.3
L-WF3	Handle consultation requests	Lawyer	UC-09	3.4
L-WF4	Work on a shared case	Lawyer	UC-09	3.5

Work-flow ID	Workflow	Actor	Related use case	Section
A-WF1	Lawyer verification	Admin	UC-10	4.2
A-WF2	Legal knowledge base management	Admin	UC-11	4.3
A-WF3	User and lawyer account management	Admin	UC-12	4.4
A-WF4	Content moderation	Admin	UC-12	4.5
A-WF5	Monitoring and audit	Admin	UC-12	4.6
S-WF1	RAG guidance pipeline	AI Services	UC-03	5.1
S-WF2	Document processing pipeline	AI Services	UC-04	5.2
S-WF3	Knowledge base ingestion pipeline	AI Services	UC-11	5.3
S-WF4	Notification dispatch	Notification Service	UC-06, UC-09	5.4

1.4 End-to-End Workflow Across Actors

Figure 1.1 shows the complete journey of one case across all actors in swimlanes. It also shows the two preconditions that the Admin must satisfy before the citizen's journey can succeed: a verified knowledge base and verified lawyers.

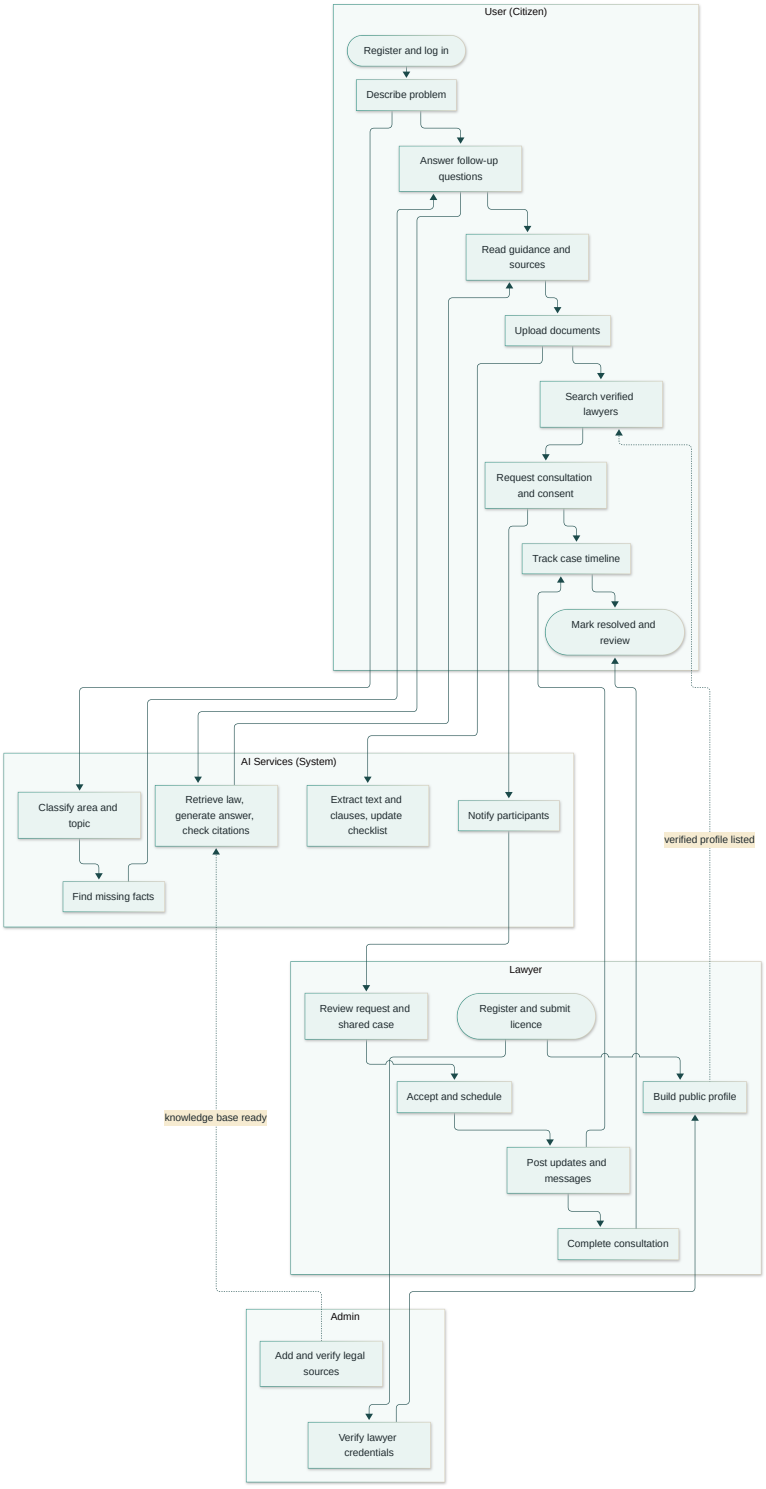


Figure 1.1: End-to-end swimlane workflow across all actors

1.5 Actor-Action Matrix

Table 1.2 summarises which actor can perform each group of actions. Detailed action catalogues with dependencies follow in Chapters 2 to 5.

Table 1.2: Actor-action matrix

Action group	User	Lawyer	Admin	AI Services
Register, log in, reset password	Yes	Yes	Log in only	No
Manage own profile	Personal profile	Public professional profile	No	No
Create and manage cases	Yes	No	View for moderation	No
Answer follow-up questions	Yes	No	No	Generates questions
Request AI legal guidance	Yes	No	No	Executes RAG pipeline
Upload documents	Yes	No	No	Extracts text and clauses
Search lawyers and view public profiles	Yes	No	Yes	No
Request consultation, give or revoke consent	Yes	No	No	No
Accept, decline or schedule consultations	No	Yes	No	No
View shared case	Owner	With active consent	Moderation only	No
Message within consultation	Yes	Yes	No	No

Action group	User	Lawyer	Admin	AI Services
Post case status updates	Mark resolved or closed	Yes	No	System events
Rate and review lawyer	Yes	No	Moderate	No
Verify lawyers	No	Submits request	Yes	No
Manage legal knowledge base	No	No	Yes	Indexes sources
Suspend or reactivate accounts	No	No	Yes	No
View statistics and audit log	No	Own dashboard	Yes	No

Chapter 2 — User (Citizen) Workflows

2.1 User Actions and Dependencies

Table 2.1: Actions of the User and their dependencies

ID	Action	Depends on	Result	Req.
U-A01	Register account	Valid, unused email	Unverified account	UFR-01
U-A02	Verify email	U-A01; email service	Active account	UFR-01
U-A03	Log in / log out	U-A02; account not suspended (A-WF3)	Access token, dashboard	UFR-02
U-A04	Reset password	Registered email	New password set	UFR-03
U-A05	Manage profile	U-A03	Profile updated	UFR-04
U-A06	Delete account	U-A03	Data removed or anonymised	UFR-05
U-A07	Create case and describe problem	U-A03	Case and chat session created	UFR-06, UFR-16
U-A08	Confirm or correct legal area and topic	U-A07; classification (S-WF1)	Area and topic fixed	UFR-07
U-A09	Answer follow-up questions	U-A08	Case status Clarified	UFR-08
U-A10	Request AI guidance	U-A09; verified, indexed sources (A-WF2)	Guidance with citations	UFR-09 to UFR-14
U-A11	Ask further questions	U-A10	Additional guidance in session	UFR-15
U-A12	Upload documents	U-A07	Document stored, text extracted (S-WF2)	UFR-17, UFR-18

ID	Action	Depends on	Result	Req.
U-A13	Review extracted clauses and checklist	U-A12 and/or U-A10	Checklist progress updated	UFR-19, UFR-20
U-A14	Search lawyers and view profile	At least one verified lawyer (A-WF1)	Lawyer selected	UFR-22, UFR-23
U-A15	Request consultation	U-A07, U-A14; lawyer accepting requests (L-WF2)	Consultation Requested	UFR-24
U-A16	Give consent to share case	U-A15	Consent record created	UFR-25
U-A17	Revoke sharing	U-A16	Lawyer access ends	UFR-26
U-A18	Message lawyer	Consultation accepted (L-WF3)	Message thread	UFR-27
U-A19	Track case progress	U-A07	Timeline shown	UFR-29
U-A20	Mark case resolved or closed	U-A07	Case Resolved or Closed	UFR-16
U-A21	Rate and review lawyer	Consultation Completed (L-WF4)	Review published	UFR-28

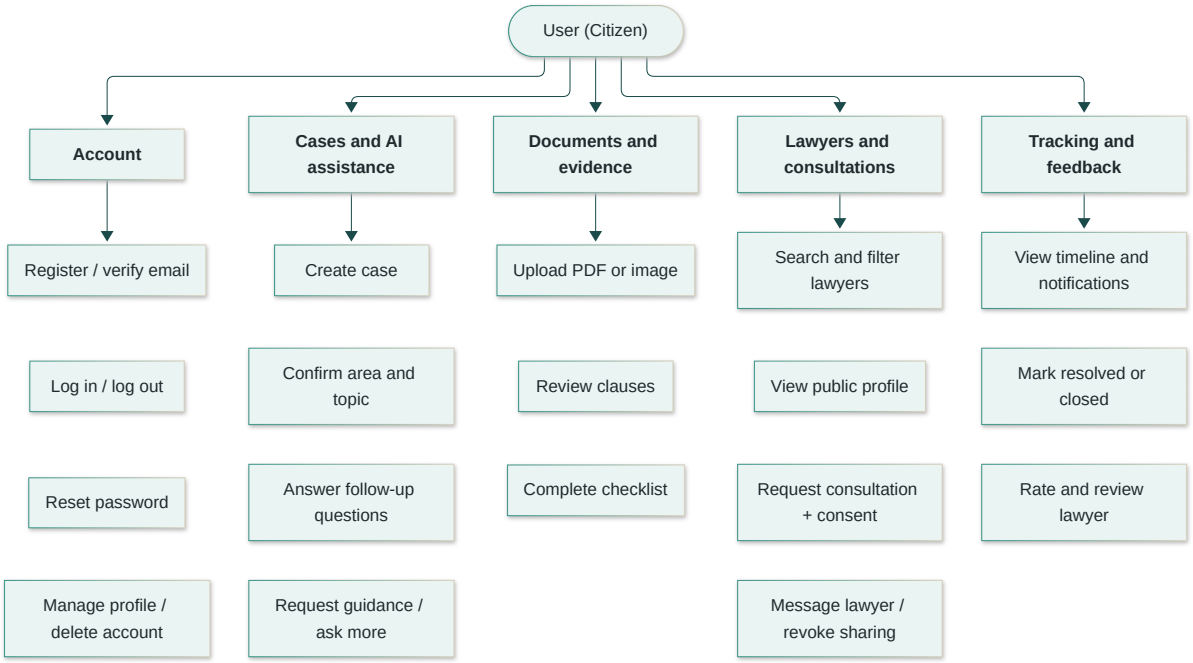


Figure 2.1: Actions available to the User

2.2 U-WF1: Account Access and Profile Management

The user registers, verifies the email address and logs in. From the dashboard the user can update the profile or request account deletion. If the account has been suspended by the Admin (A-WF3), login is refused.

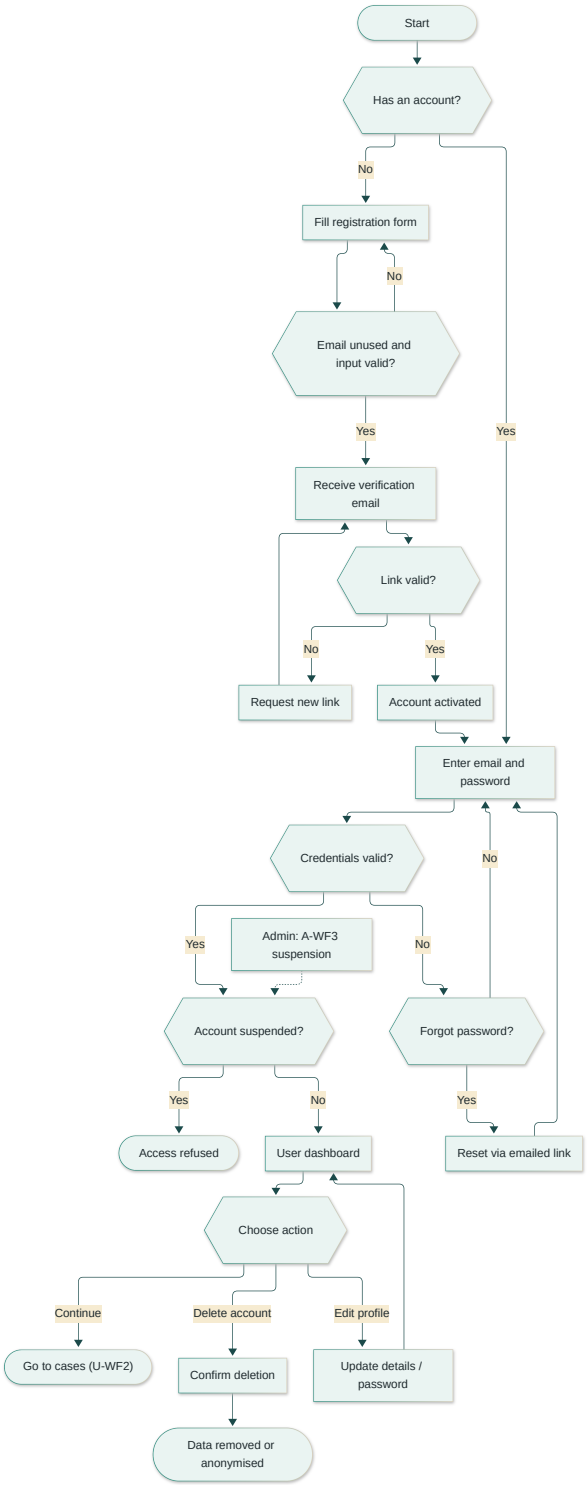


Figure 2.2: U-WF1 Account access and profile management

2.3 U-WF2: Create Case and Clarify Facts

The user describes the problem in free text. The system classifies the legal area and topic, and the user can correct this. The system then asks follow-up questions until all required facts for the topic are known. A question outside property and family law ends the workflow with a referral to the lawyer directory.

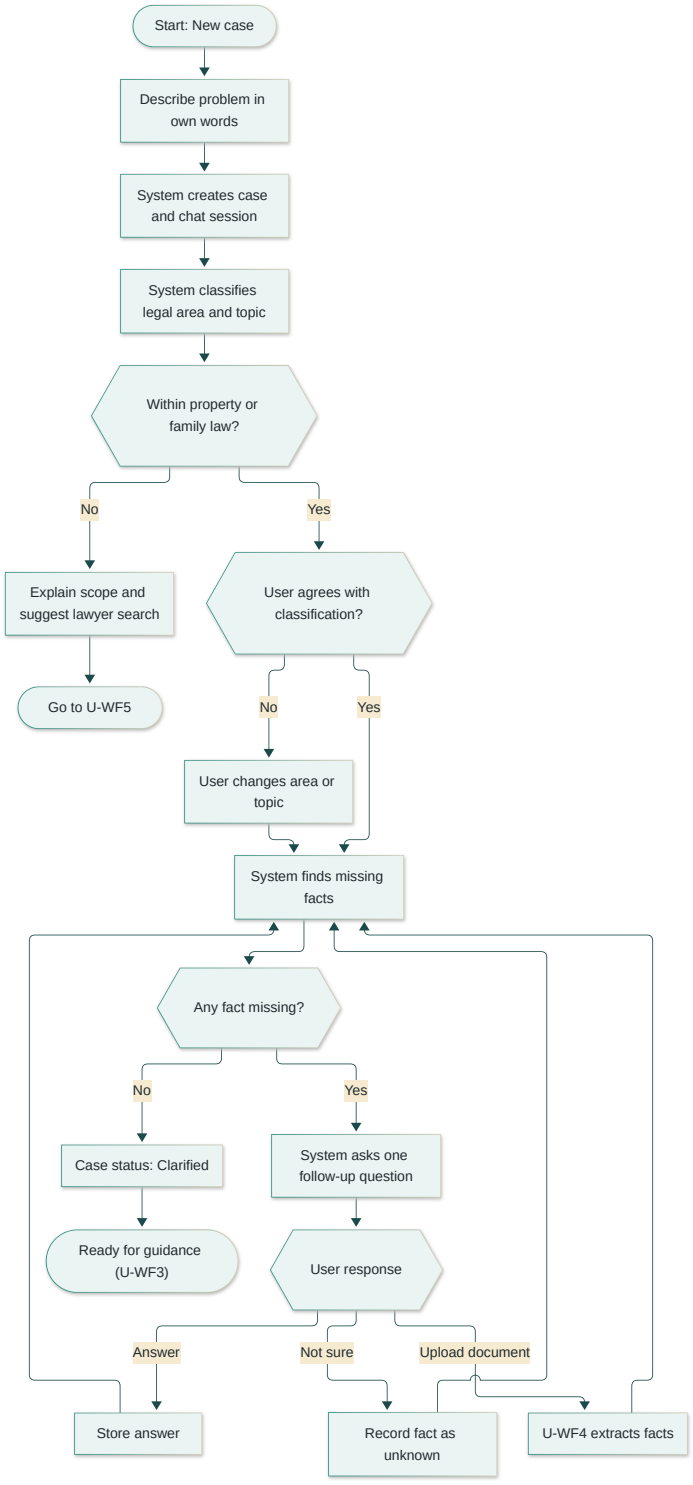


Figure 2.3: U-WF2 Create case and clarify facts

2.4 U-WF3: Obtain AI Legal Guidance

Guidance can be requested only for a clarified case and depends on the knowledge base prepared by the Admin (A-WF2). The automated steps are detailed in S-WF1. The user either receives a cited answer or an honest “no verified source” response that points to the lawyer directory.

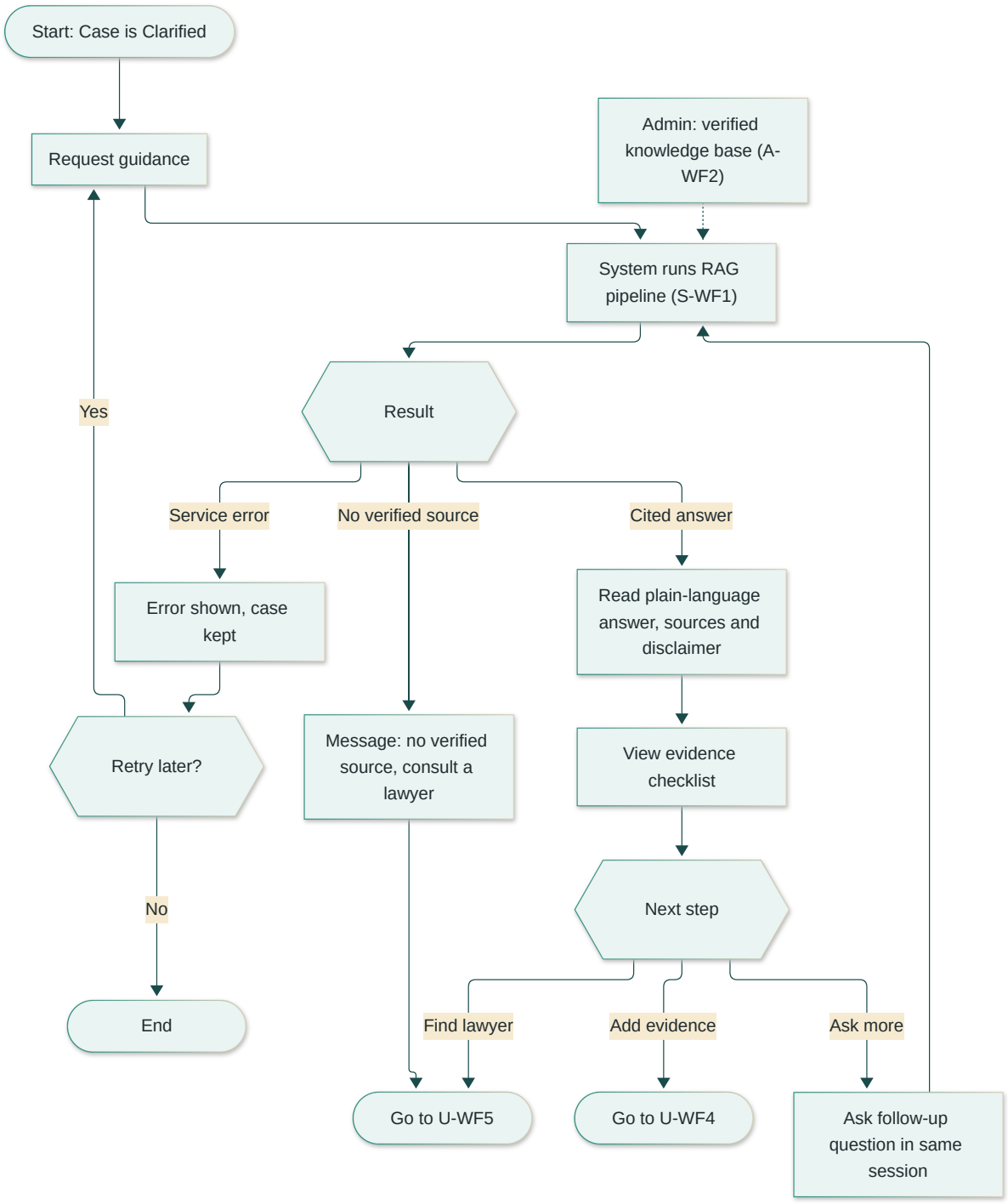


Figure 2.4: U-WF3 Obtain AI legal guidance

2.5 U-WF4: Upload Documents and Complete Evidence Checklist

Documents can be uploaded at any time after a case exists. Text extraction (PyMuPDF or OCR) and clause extraction are automated in S-WF2. Uploaded documents that match checklist items mark those items as available.

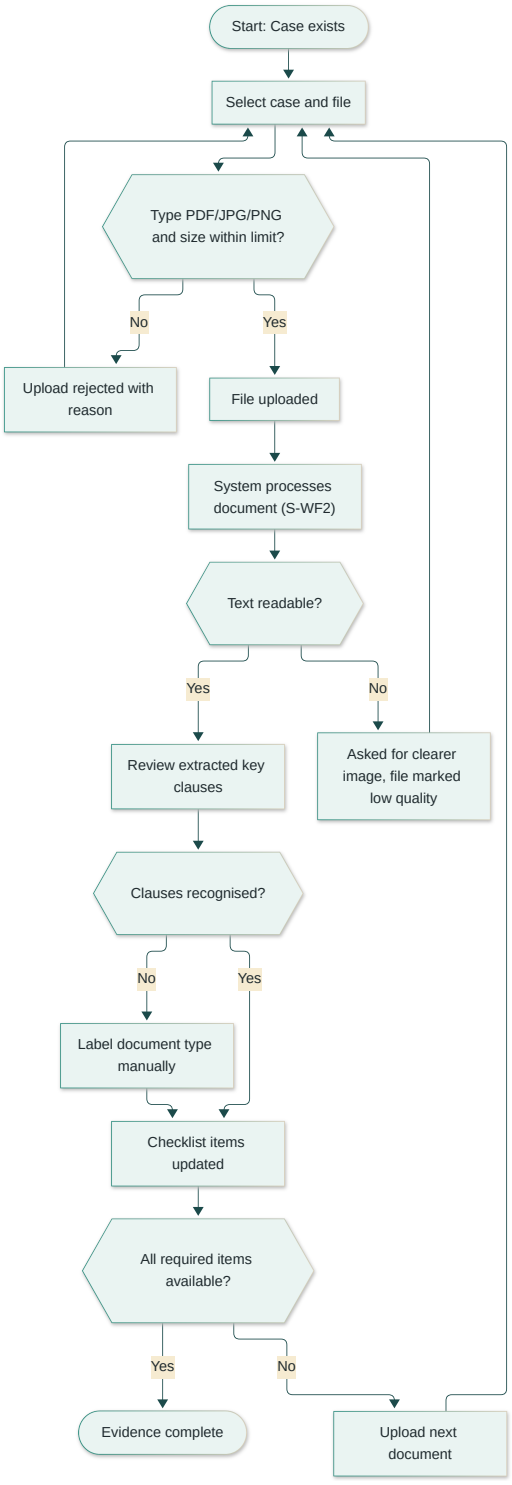


Figure 2.5: U-WF4 Upload documents and complete evidence checklist

2.6 U-WF5: Find a Lawyer and Request a Consultation

The directory shows only lawyers verified by the Admin (A-WF1) whose profiles are published (L-WF2). A consultation is created only after explicit consent, and the request is delivered to the lawyer, who handles it in L-WF3.

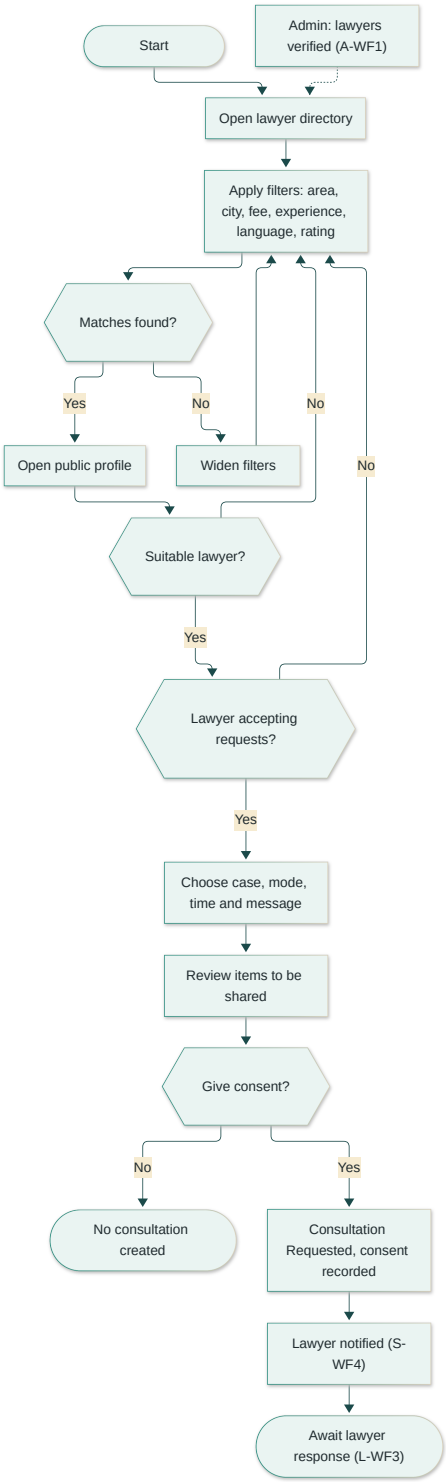


Figure 2.6: U-WF5 Find a lawyer and request a consultation

2.7 U-WF6: Track Case, Communicate and Review

The user follows the case timeline, receives notifications of lawyer actions, can message the lawyer within an accepted consultation, can revoke sharing at any time, and reviews the lawyer after the consultation is completed.

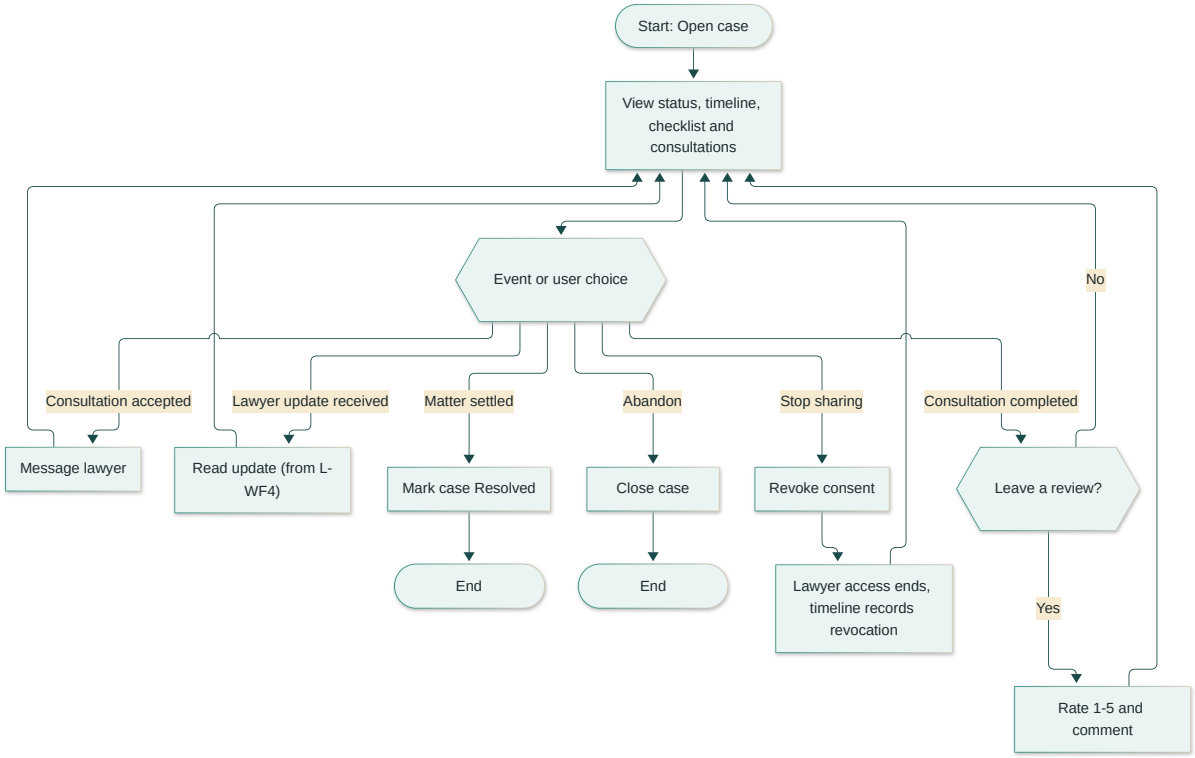


Figure 2.7: U-WF6 Track case, communicate and review

Chapter 3 — Lawyer Workflows

3.1 Lawyer Actions and Dependencies

Table 3.1: Actions of the Lawyer and their dependencies

ID	Action	Depends on	Result	Req.
L-A01	Register as lawyer with bar details and licence	Valid email	Account pending verification	LFR-01
L-A02	Log in / log out	Verified email; not suspended (A-WF3)	Lawyer dashboard	LFR-02
L-A03	Build or edit public profile	L-A02	Profile saved (preview)	LFR-03
L-A04	Submit or re-submit verification request	L-A01	Request in Admin queue	LFR-01
L-A05	Profile published in directory	Admin approval (A-WF1)	Verified badge, listed	LFR-04
L-A06	Set availability / pause requests	L-A05	Accepting or paused	LFR-05
L-A07	View consultation requests	L-A05; user request with consent (U-WF5)	Request list	LFR-06
L-A08	Accept, decline or propose another time	L-A07	Consultation Scheduled, Declined or Rescheduling	LFR-07
L-A09	Access shared case	Active consent (U-A16, not revoked)	Read-only case view	LFR-08
L-A10	Message user	Consultation accepted	Message thread	LFR-09
L-A11	Post case status update	Consultation accepted; active consent	Timeline entry, user notified	LFR-10

ID	Action	Depends on	Result	Req.
L-A12	Mark consultation completed	Consultation Scheduled	Review enabled for user	LFR-11
L-A13	View dashboard and rating	L-A02	Statistics	LFR-12

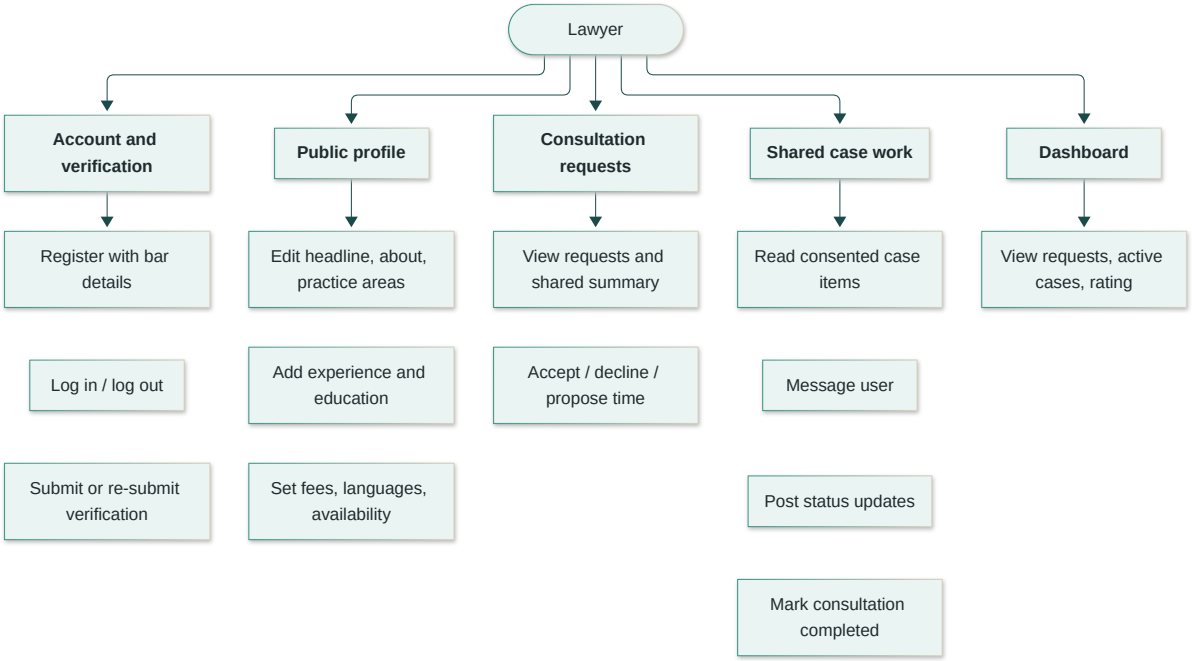


Figure 3.1: Actions available to the Lawyer

3.2 L-WF1: Registration and Verification

A lawyer can log in after email verification but is not listed in the directory until the Admin approves the submitted credentials (A-WF1). A rejected lawyer can correct the details and re-submit.

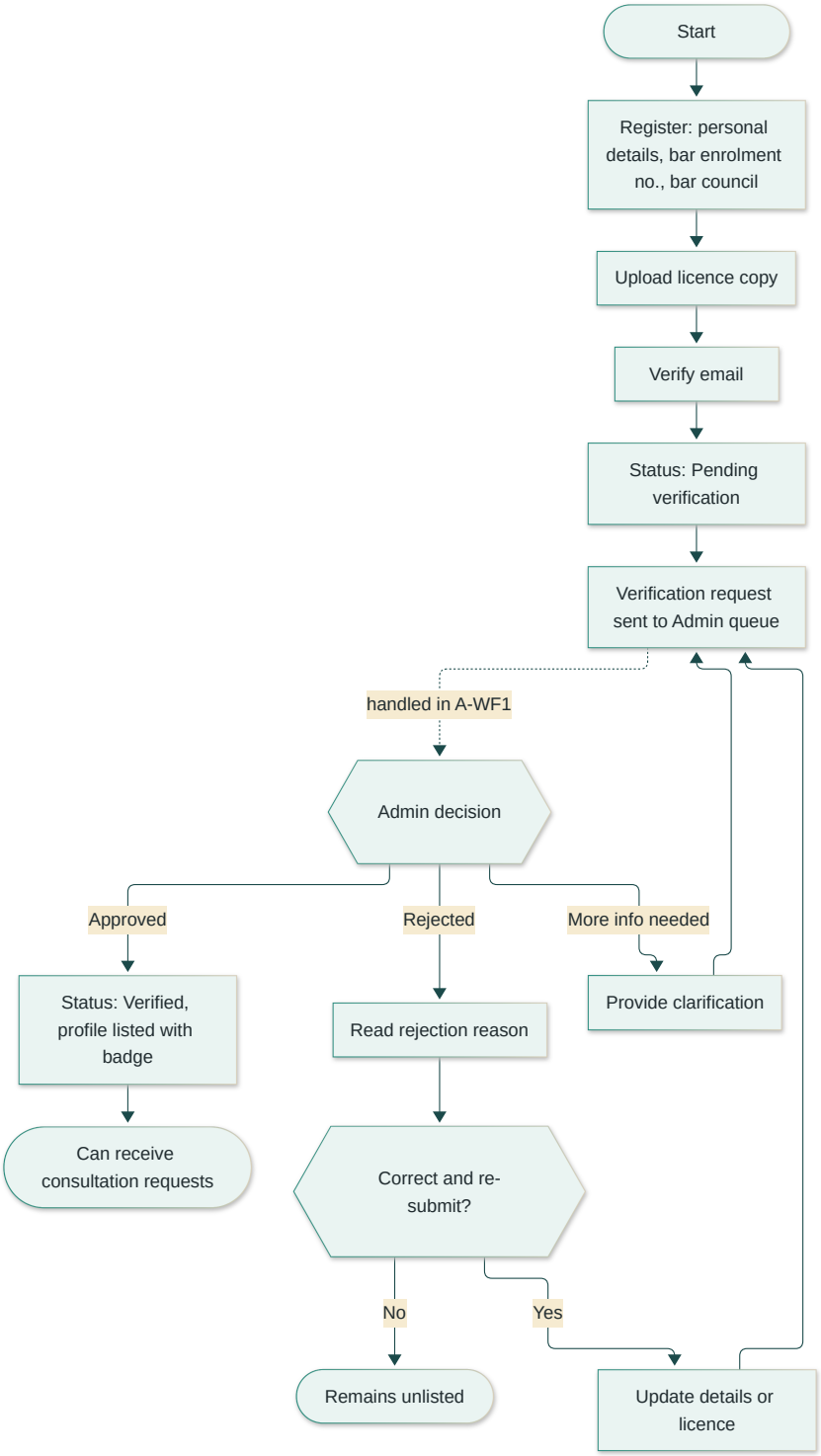


Figure 3.2: L-WF1 Registration and verification

3.3 L-WF2: Public Profile and Availability Management

The profile is a LinkedIn-style public page. Edits to ordinary profile fields are published immediately for a verified lawyer. Edits to bar enrolment details return the profile to pending verification.

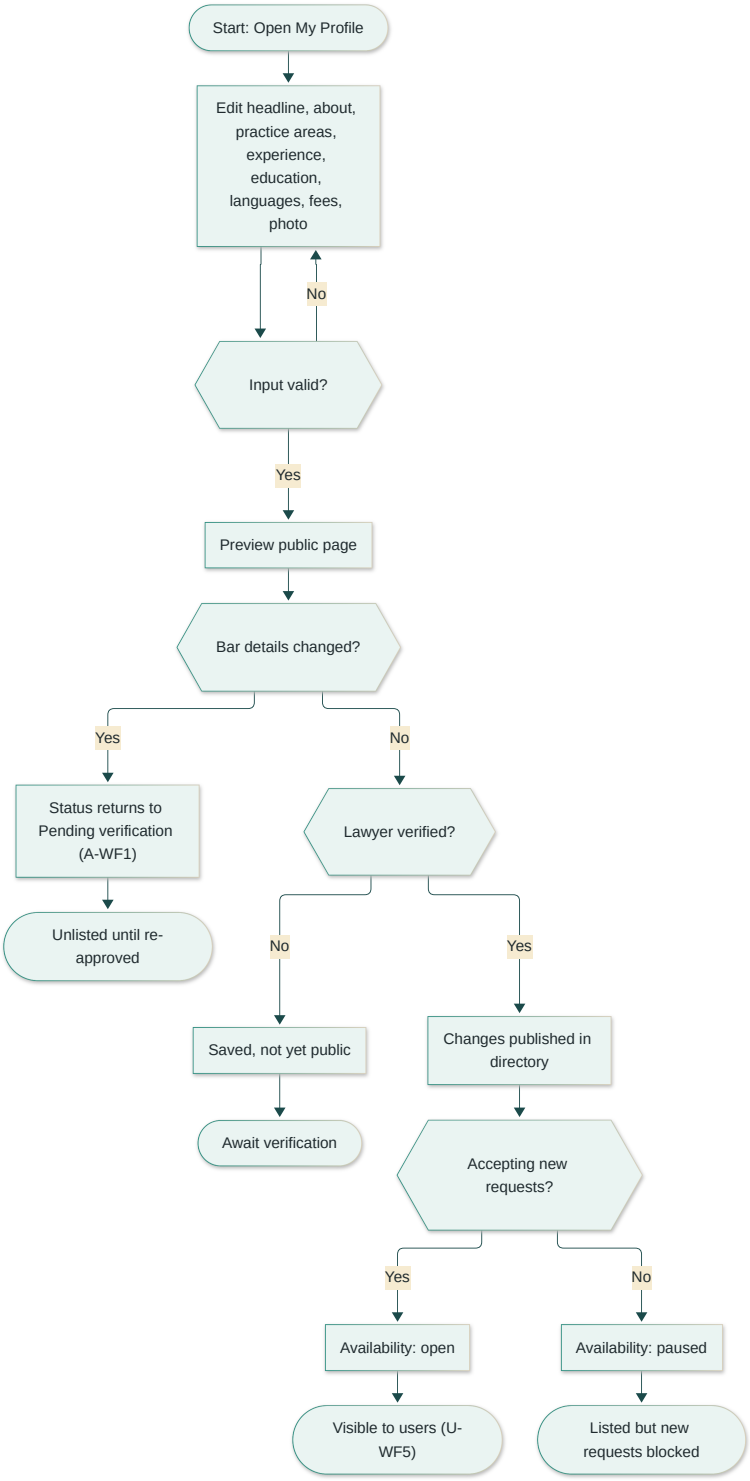


Figure 3.3: L-WF2 Public profile and availability management

3.4 L-WF3: Handle Consultation Requests

Requests arrive only from users who gave consent (U-WF5). Before every read of the shared case, the system checks that consent is still active.

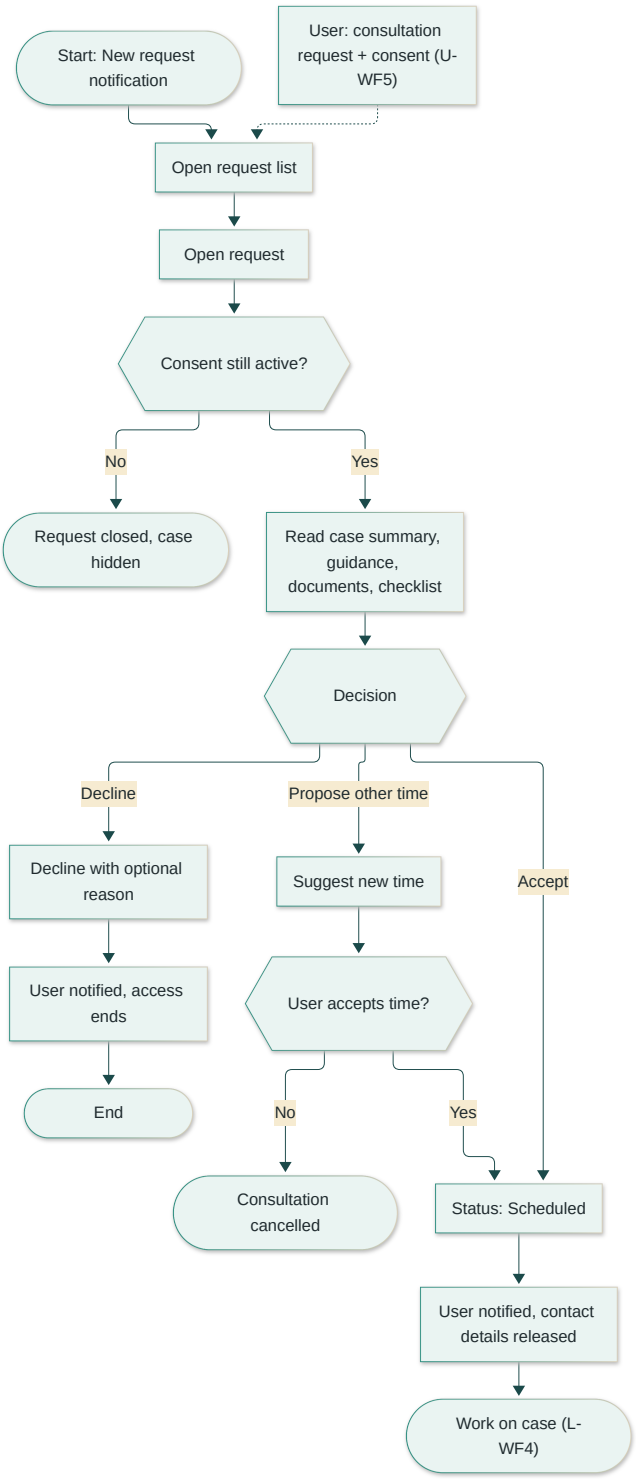


Figure 3.4: L-WF3 Handle consultation requests

3.5 L-WF4: Work on a Shared Case

While the consultation is active, the lawyer exchanges messages and posts status updates that appear on the user's timeline (U-WF6). Completing the consultation enables the user's review.

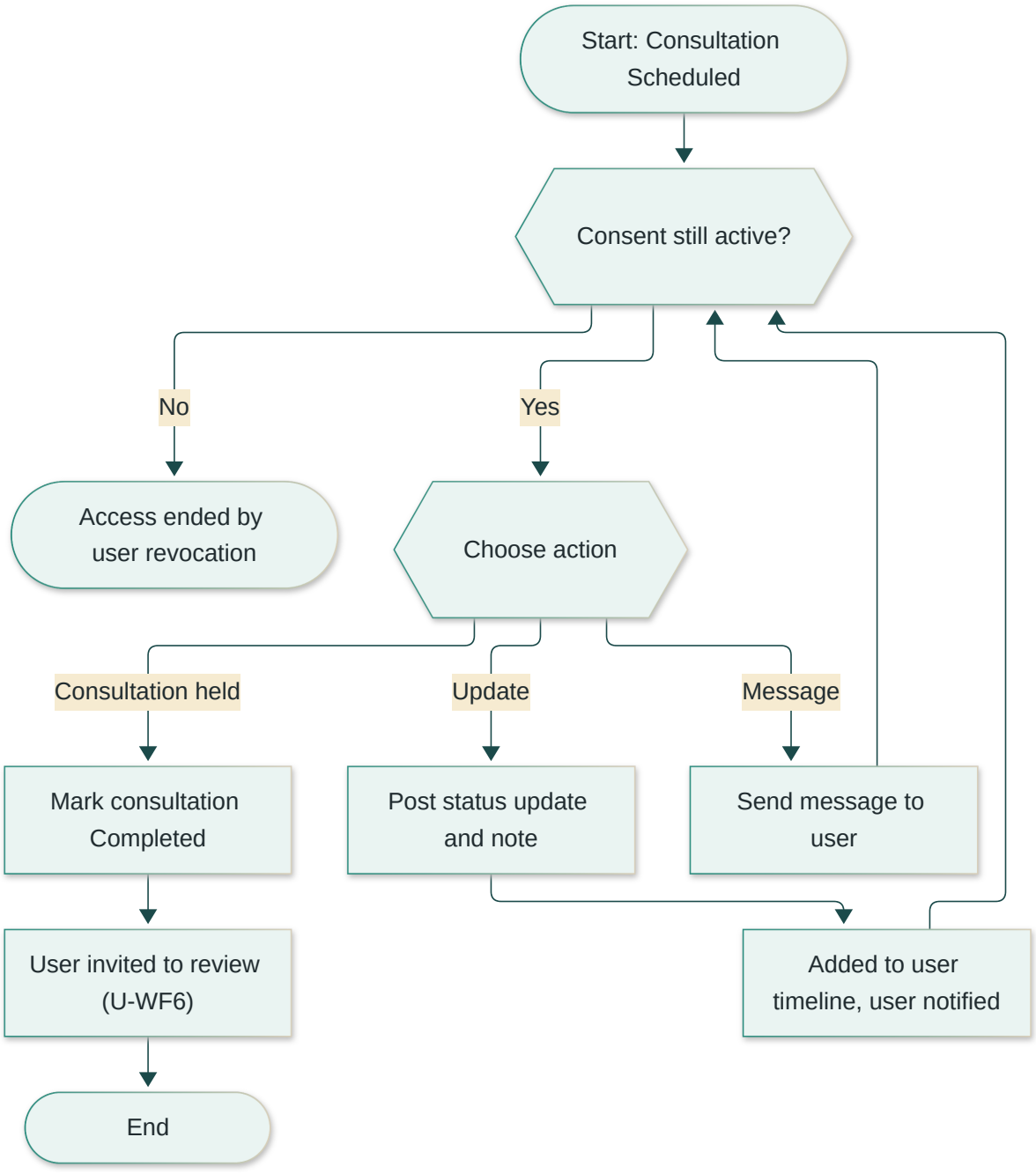


Figure 3.5: L-WF4 Work on a shared case

Chapter 4 — Admin Workflows

4.1 Admin Actions and Dependencies

Table 4.1: Actions of the Admin and their dependencies

ID	Action	Depends on	Result	Req.
A-A01	Log in to Admin portal	Administrator role	Admin dashboard	AFR-01
A-A02	Review verification queue	Pending requests (L-WF1)	Request details and licence	AFR-02
A-A03	Approve, reject or request information	A-A02; external check with bar council records	Lawyer Verified / Rejected / Pending	AFR-02
A-A04	Add legal source with metadata	Official text available	Source Processing	AFR-05
A-A05	Review indexed chunks and verify source	A-A04; ingestion done (S-WF3)	Source Verified and retrievable	AFR-05, AFR-06
A-A06	Re-index amended source / retire repealed source	A-A05	Updated or excluded from retrieval	AFR-07
A-A07	Search, view, suspend, reactivate, remove accounts	A-A01	Account status changed	AFR-03
A-A08	Moderate reported profiles and reviews	Reports from users	Content kept or removed	AFR-04
A-A09	View platform statistics	A-A01	Dashboard metrics	AFR-08
A-A10	View and filter audit log	Logged admin actions	Audit trail	AFR-09

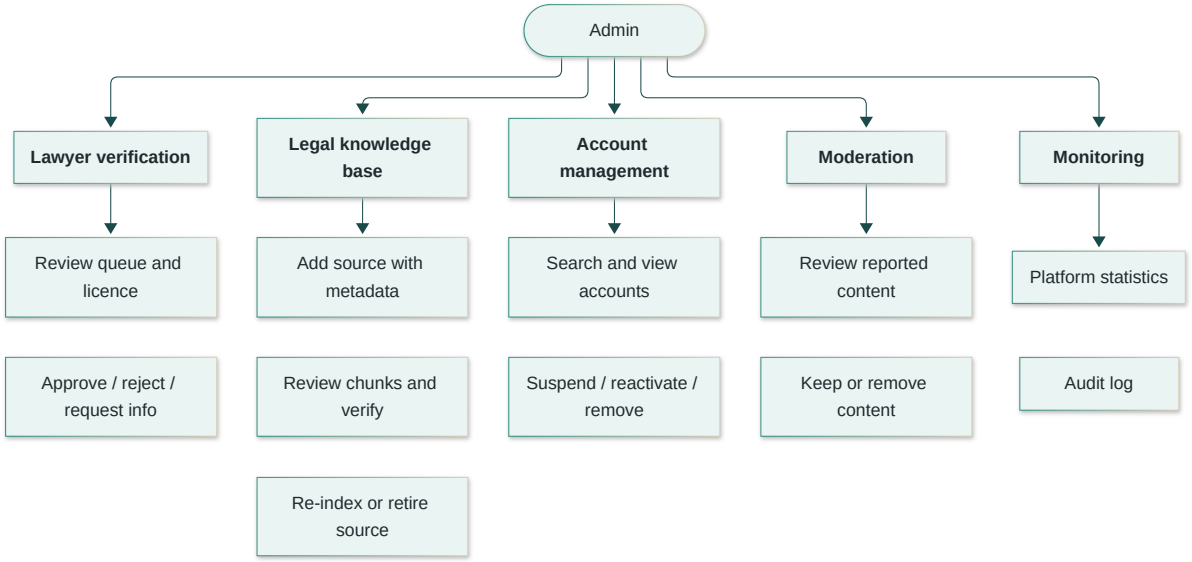


Figure 4.1: Actions available to the Admin

4.2 A-WF1: Lawyer Verification

The Admin checks each lawyer's bar enrolment details against the issuing bar council's records, which is done outside the system, and records the decision. The decision directly controls whether the lawyer appears in the user's directory (U-WF5).

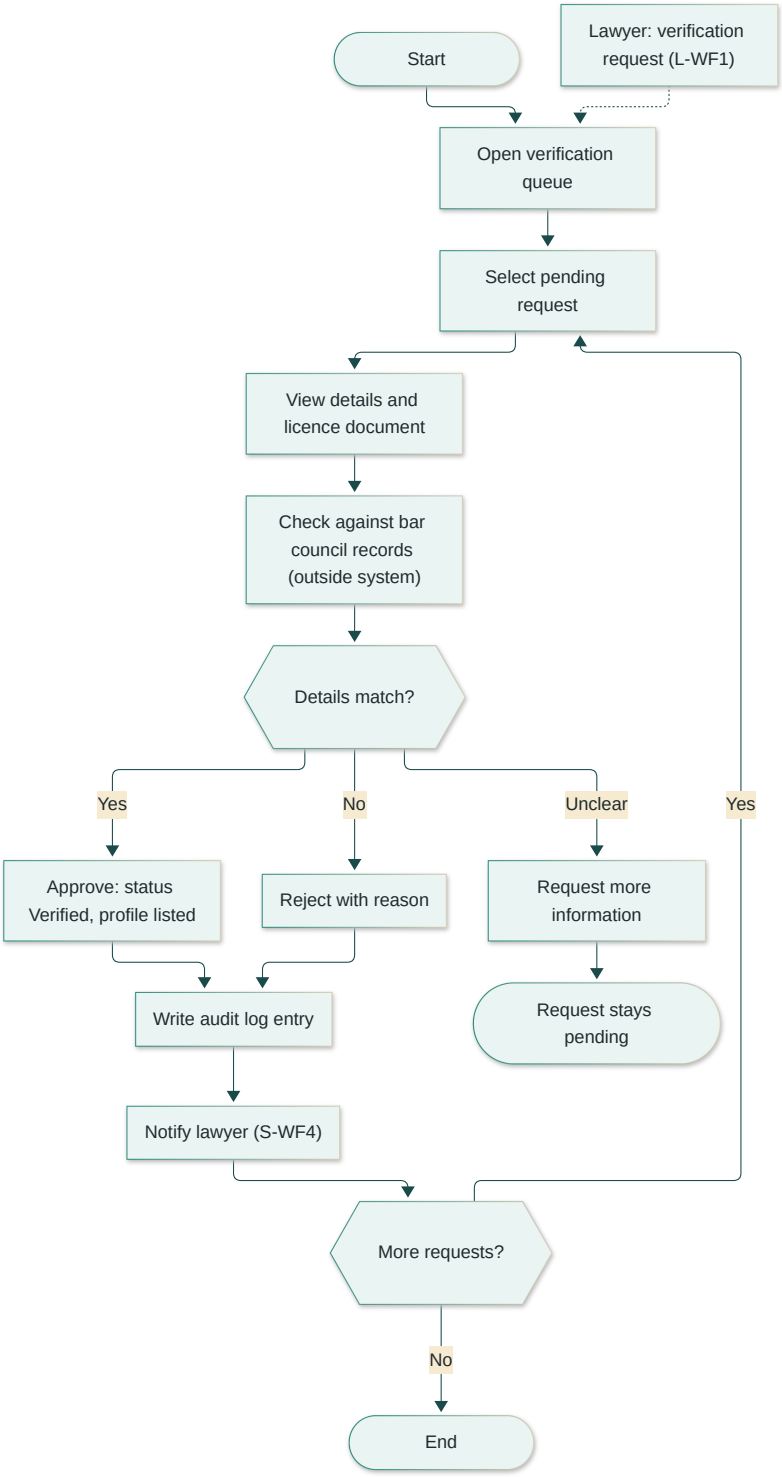


Figure 4.2: A-WF1 Lawyer verification

4.3 A-WF2: Legal Knowledge Base Management

Only sources that the Admin has reviewed and verified are retrievable by the AI assistant. This is the key control that keeps guidance grounded in verified Pakistani law.

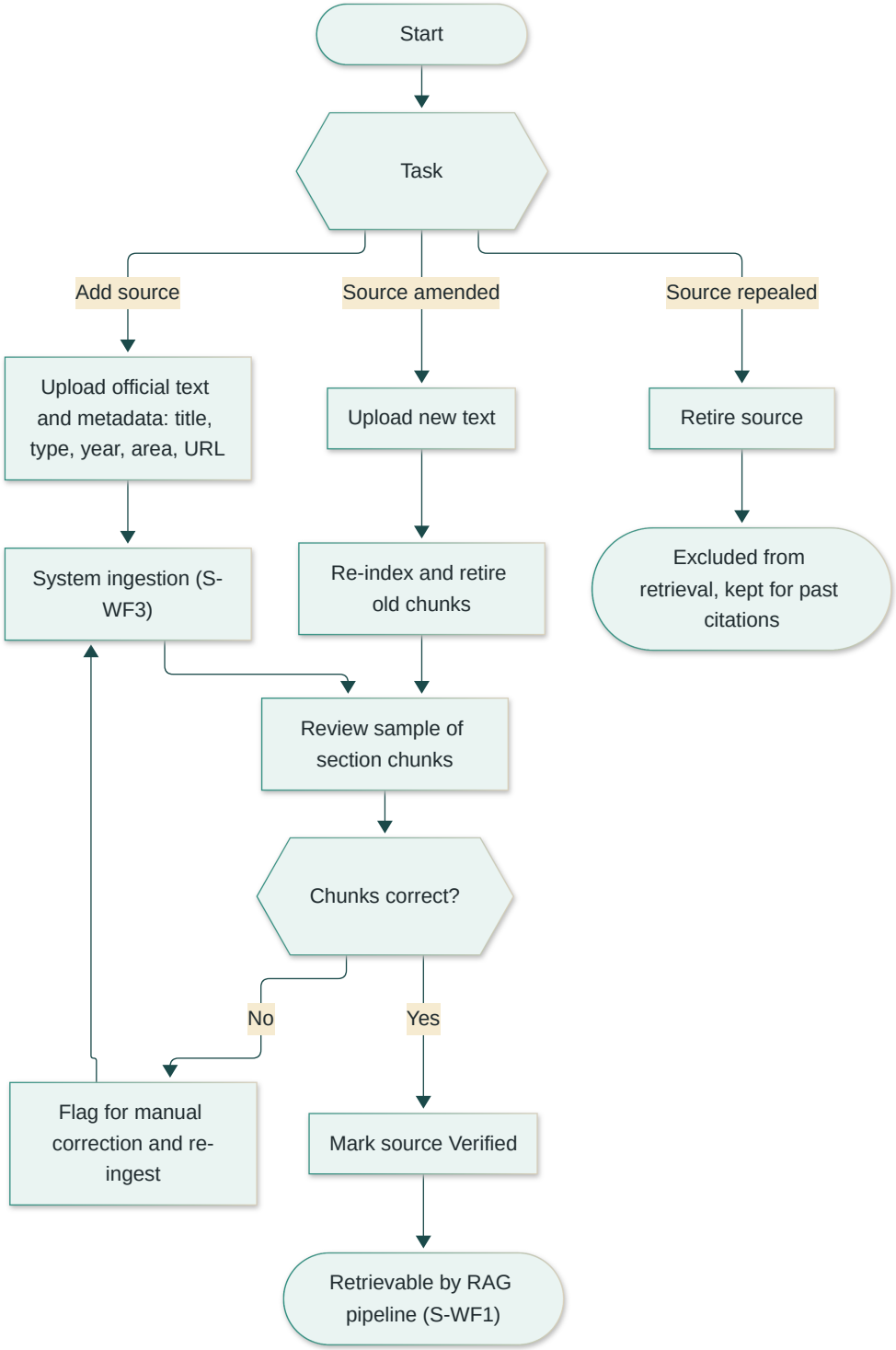


Figure 4.3: A-WF2 Legal knowledge base management

4.4 A-WF3: User and Lawyer Account Management

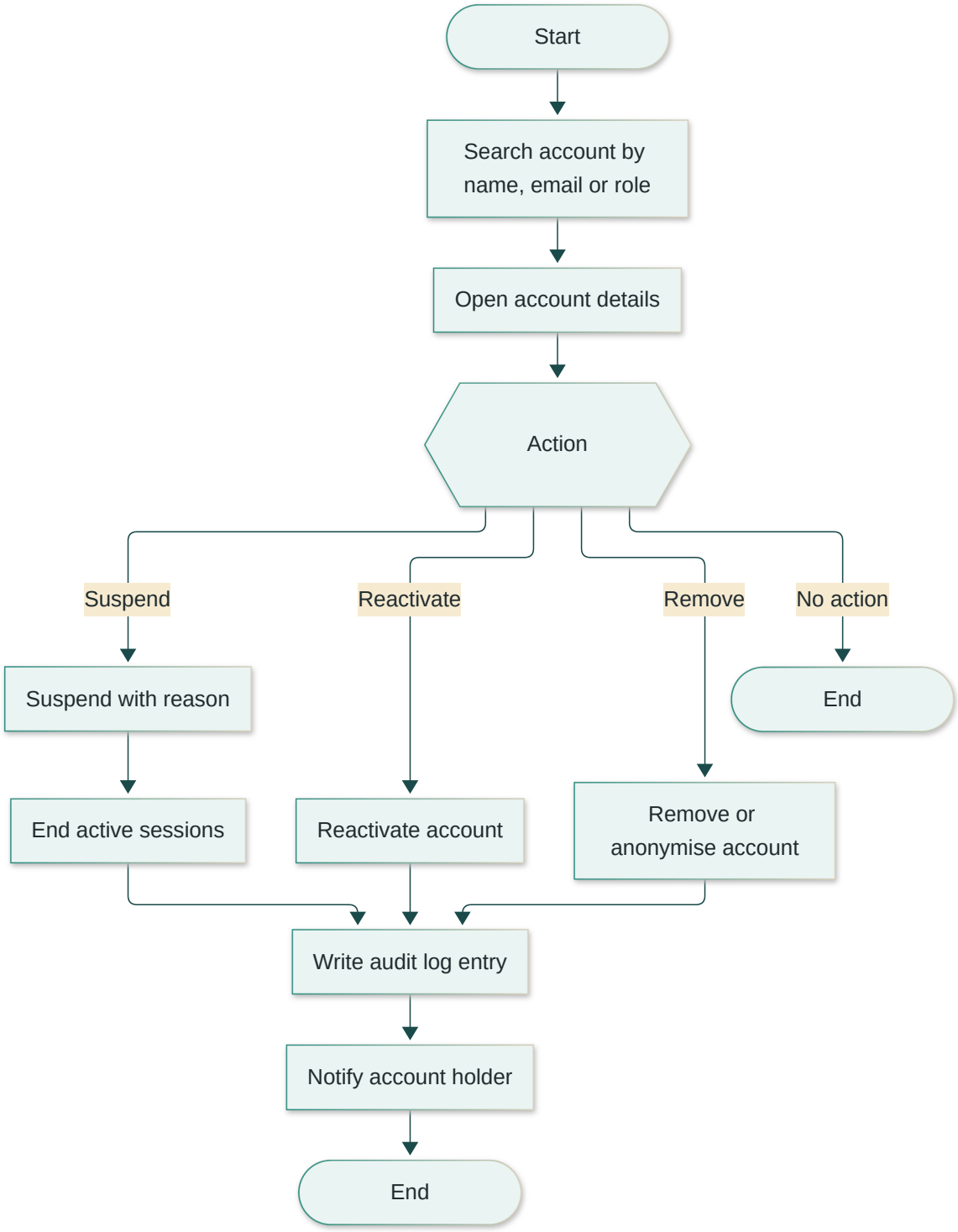


Figure 4.4: A-WF3 User and lawyer account management

Suspension takes effect immediately in U-WF1 and L-WF1: a suspended account cannot log in, and a suspended lawyer is removed from the directory.

4.5 A-WF4: Content Moderation

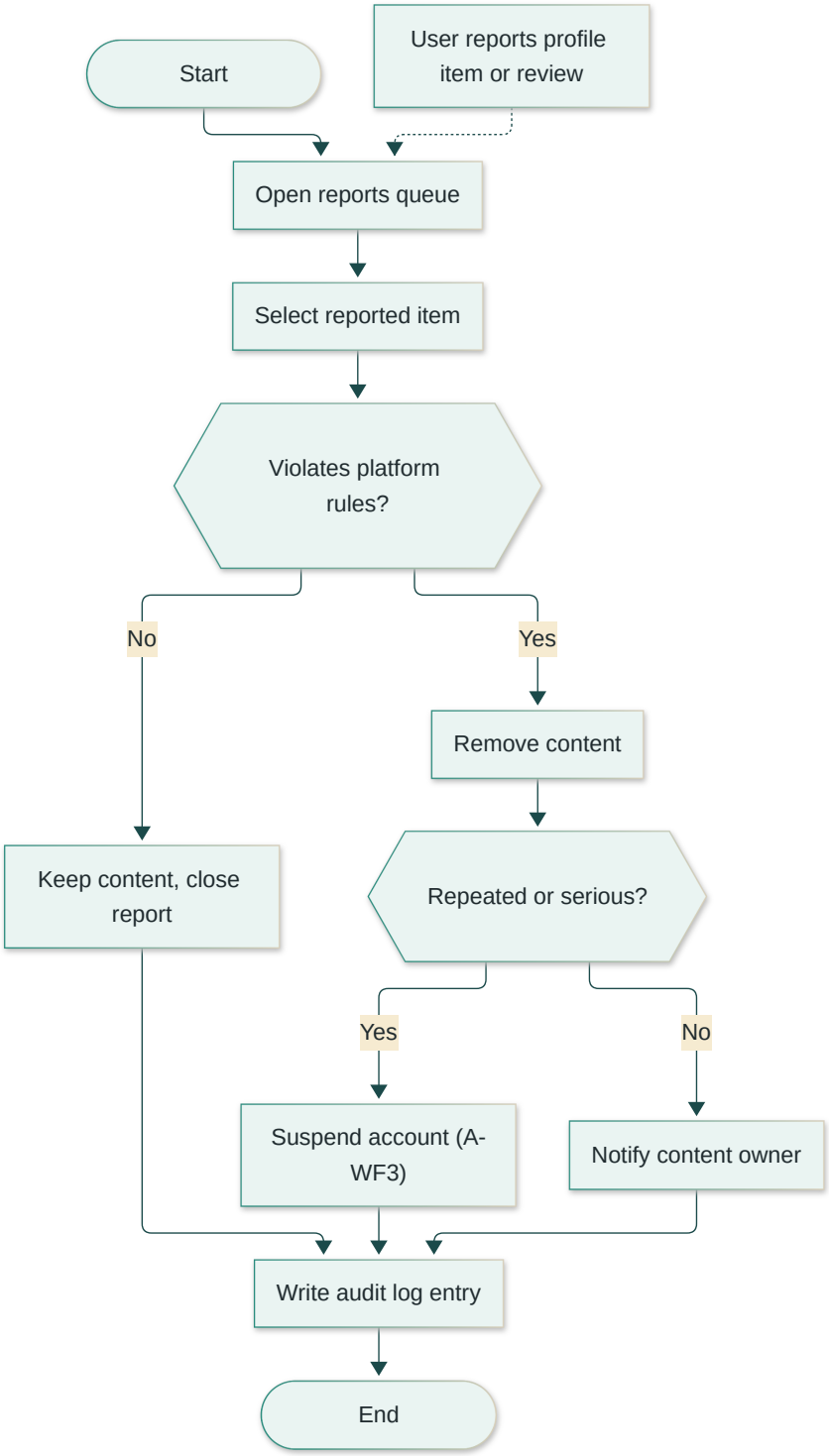


Figure 4.5: A-WF4 Content moderation

4.6 A-WF5: Monitoring and Audit

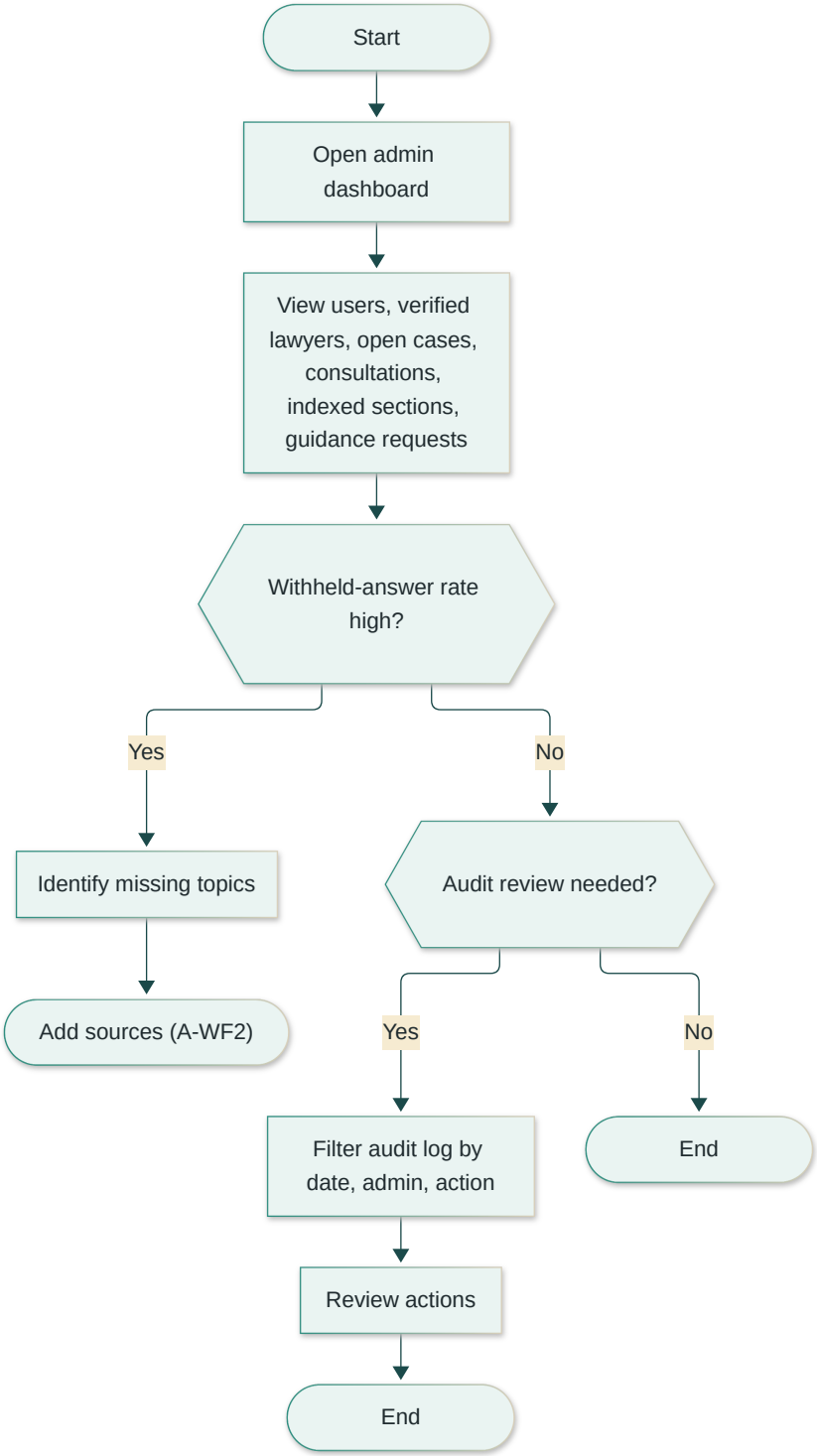


Figure 4.6: A-WF5 Monitoring and audit

A high rate of withheld ("no verified source") answers signals gaps in the knowledge base, which the Admin closes by adding sources in A-WF2.

Chapter 5 — System (AI Services) Workflows

The system workflows run automatically when a human actor triggers them. They are shown as swimlanes across the backend components.

5.1 S-WF1: RAG Guidance Pipeline

Triggered by U-A10 and U-A11. The LLM receives only the retrieved sections, and no answer is shown unless its citations pass verification.

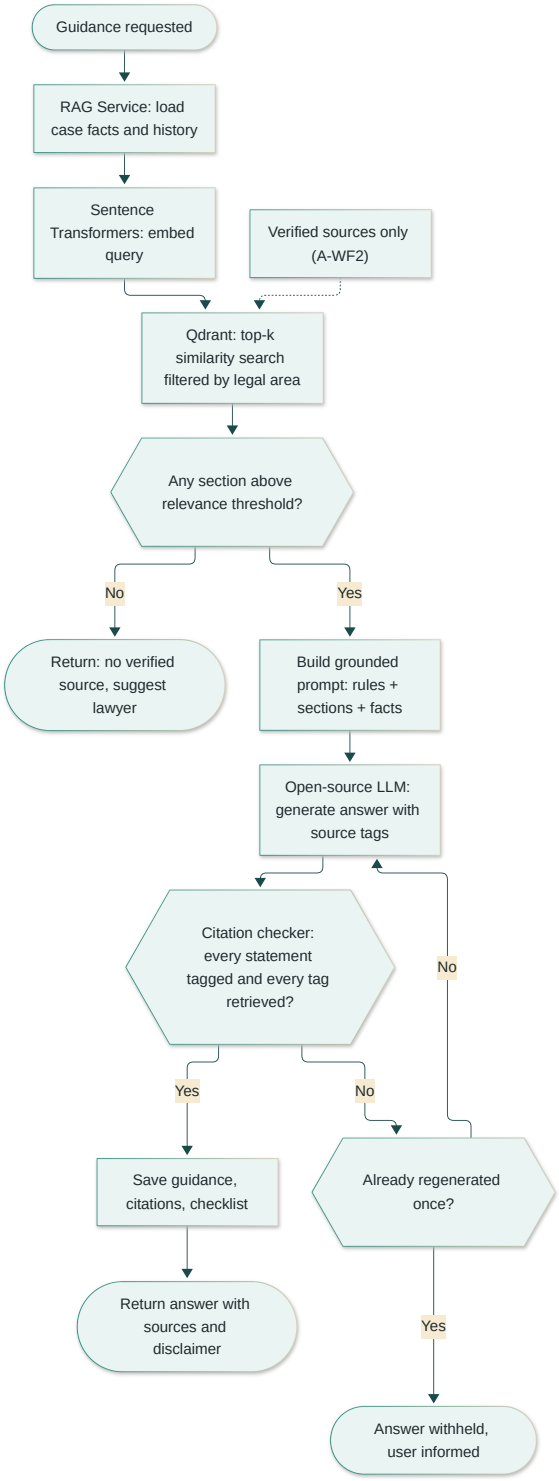


Figure 5.1: S-WF1 RAG guidance pipeline

5.2 S-WF2: Document Processing Pipeline

Triggered by U-A12. Runs asynchronously so the user is not blocked.

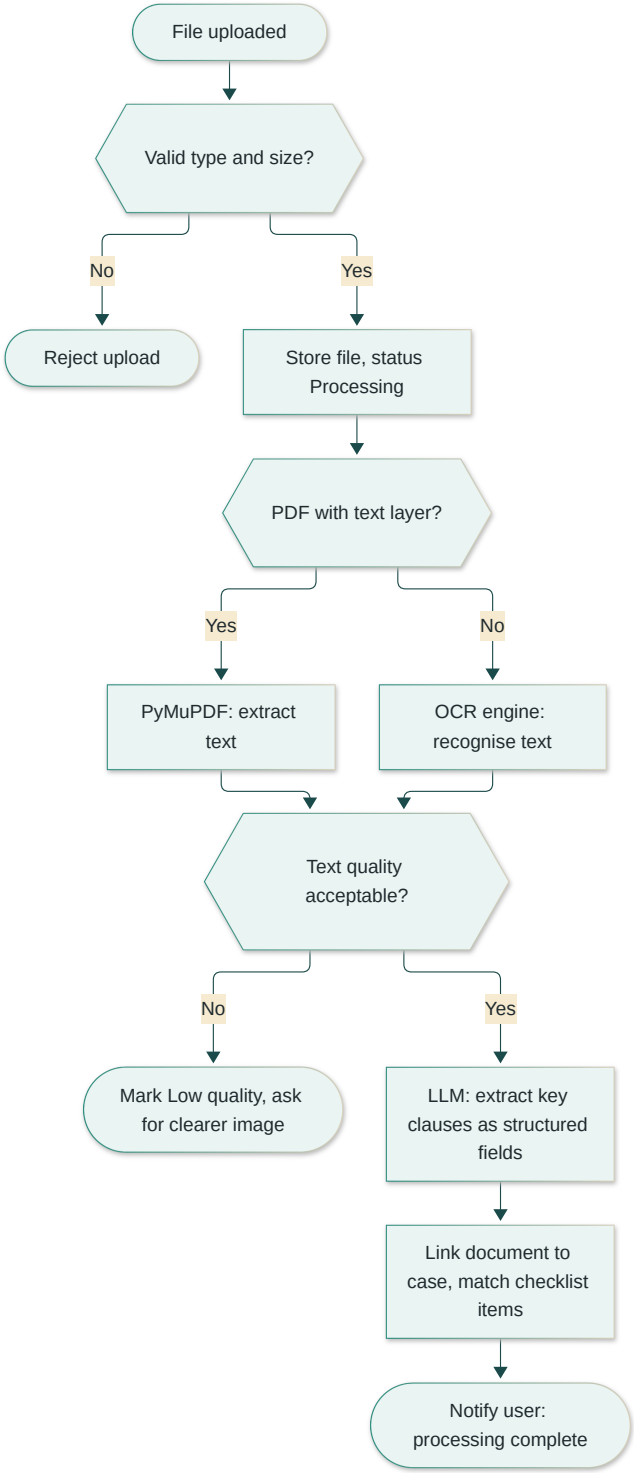


Figure 5.2: S-WF2 Document processing pipeline

5.3 S-WF3: Knowledge Base Ingestion Pipeline

Triggered by A-A04 and A-A06.

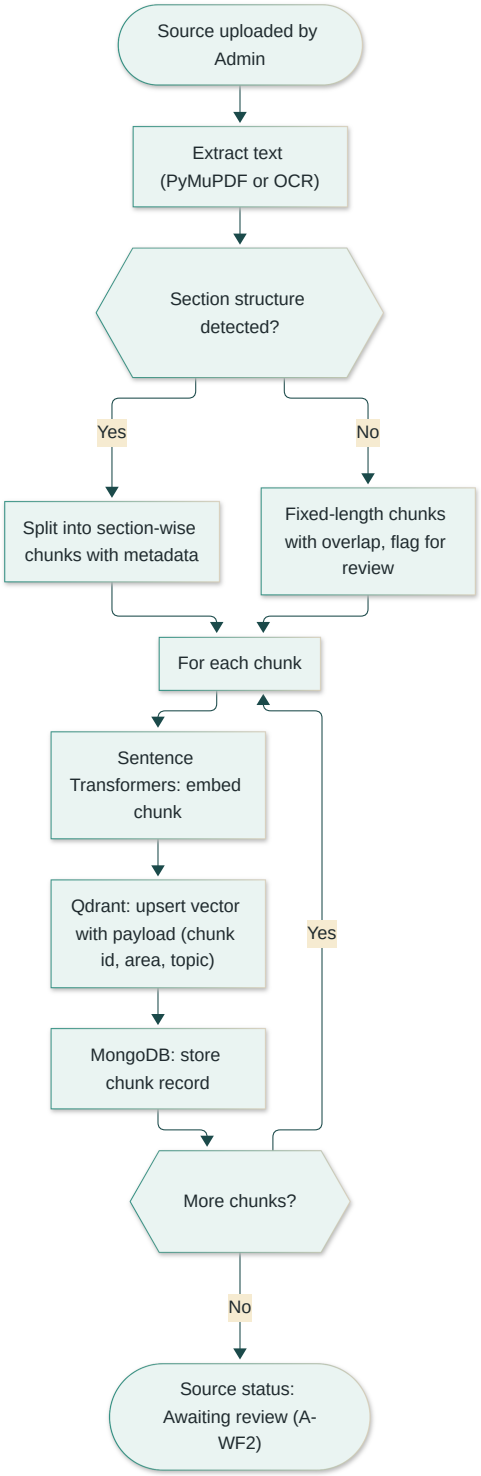


Figure 5.3: S-WF3 Knowledge base ingestion pipeline

5.4 S-WF4: Notification Dispatch

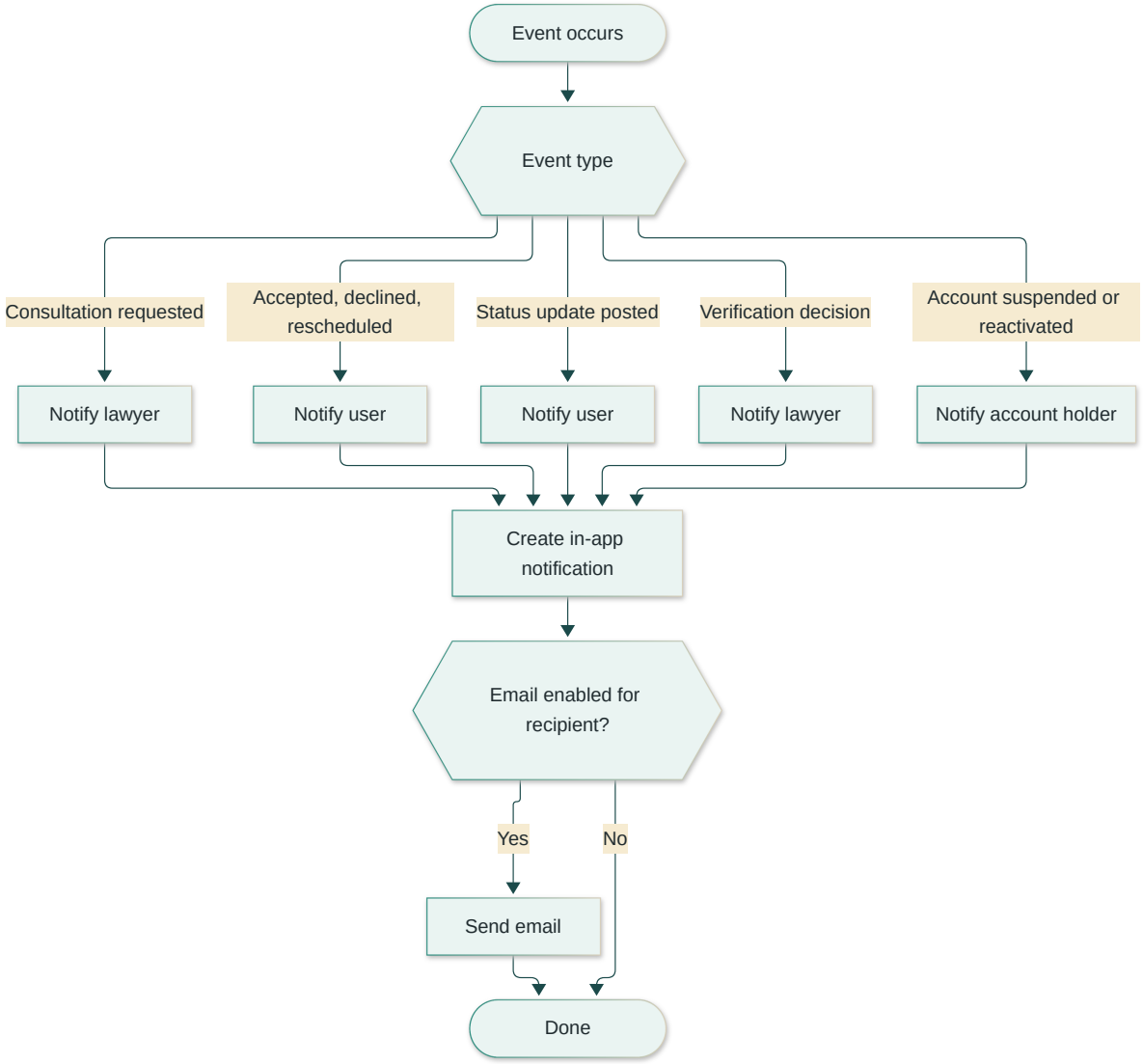


Figure 5.4: S-WF4 Notification dispatch

Chapter 6 — Cross-Actor Dependencies and Status Lifecycles

6.1 Dependency Graph

Figure 6.1 shows which workflow outcomes must exist before another workflow can proceed. Arrows point from the prerequisite to the dependent step. Two chains start with the Admin. The **knowledge chain** (A-WF2 to S-WF3 to S-WF1) makes AI guidance possible. The **trust chain** (L-WF1 to A-WF1 to L-WF2 to U-WF5) makes lawyers discoverable. The consultation then depends on both the user's case and the user's consent.

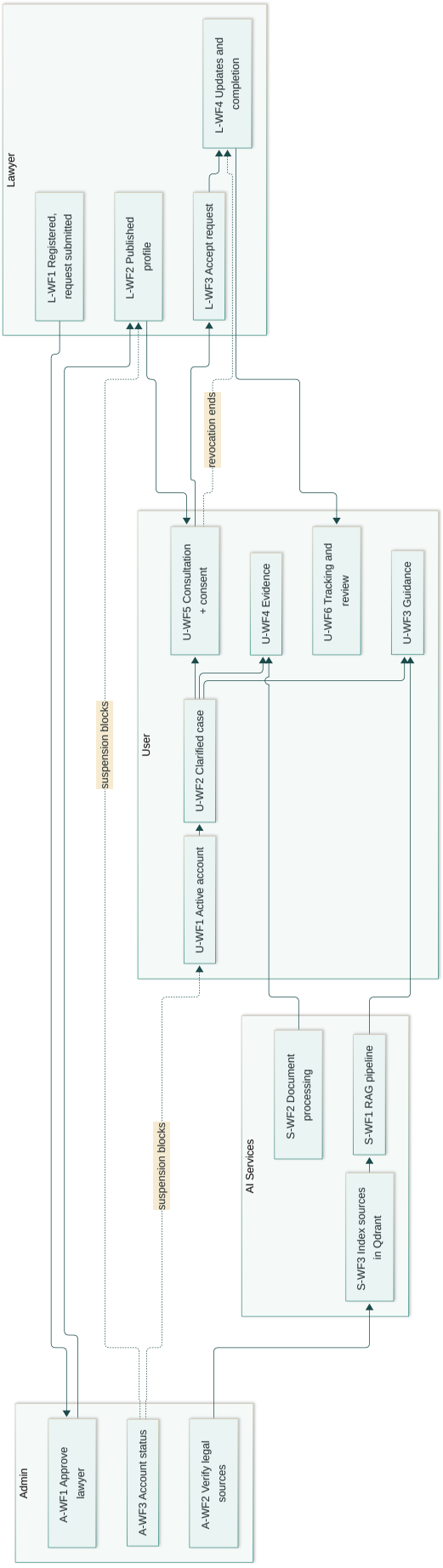


Figure 6.1: Dependency graph between workflows and actors

Table 6.1: Key dependencies

Dependent step	Depends on	Provided by	If dependency is missing
AI guidance (U-WF3)	Verified and indexed legal sources	Admin (A-WF2) and System (S-WF3)	System returns "no verified source" and suggests a lawyer
AI guidance (U-WF3)	Case status Clarified	User (U-WF2)	Guidance is not offered until facts are complete
Evidence checklist (U-WF4)	Guidance generated or document uploaded	System (S-WF1, S-WF2)	Checklist is empty
Lawyer appears in directory (U-WF5)	Lawyer verification approved	Admin (A-WF1)	Lawyer is not listed
Consultation request (U-WF5)	Existing case, verified lawyer accepting requests	User, Lawyer (L-WF2)	Request button disabled
Lawyer reads case (L-WF3, L-WF4)	Active consent	User (U-WF5)	Access refused, request closed
Contact details released	Consultation accepted	Lawyer (L-WF3)	Contact details hidden
User messages lawyer (U-WF6)	Consultation accepted	Lawyer (L-WF3)	Messaging unavailable
Review (U-WF6)	Consultation Completed	Lawyer (L-WF4)	Review not allowed
Login (U-WF1, L-WF1)	Account not suspended	Admin (A-WF3)	Access refused
Notifications	Triggering event	System (S-WF4)	No notification sent

6.2 Case Status Lifecycle

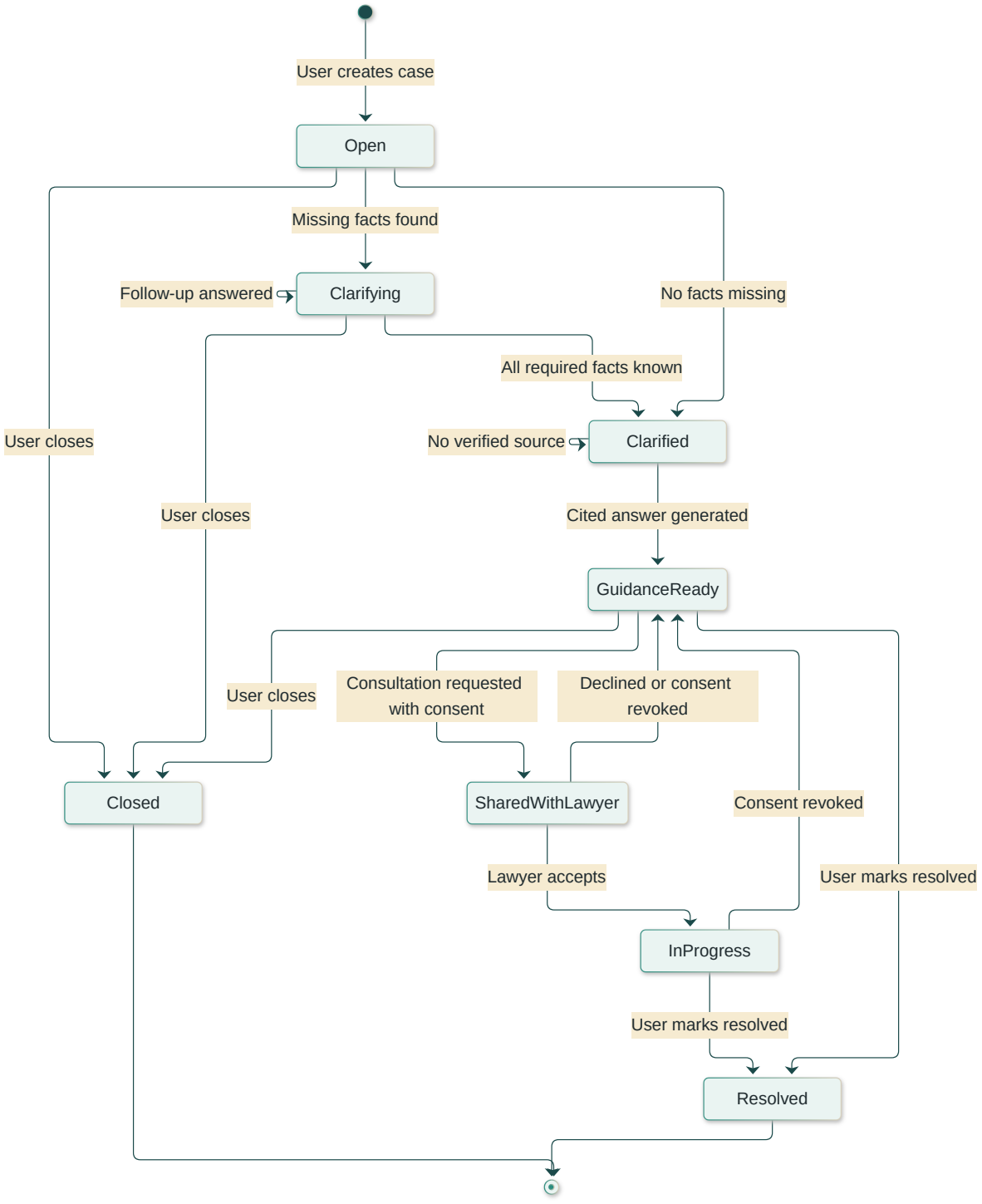


Figure 6.2: Case status lifecycle

6.3 Consultation Status Lifecycle

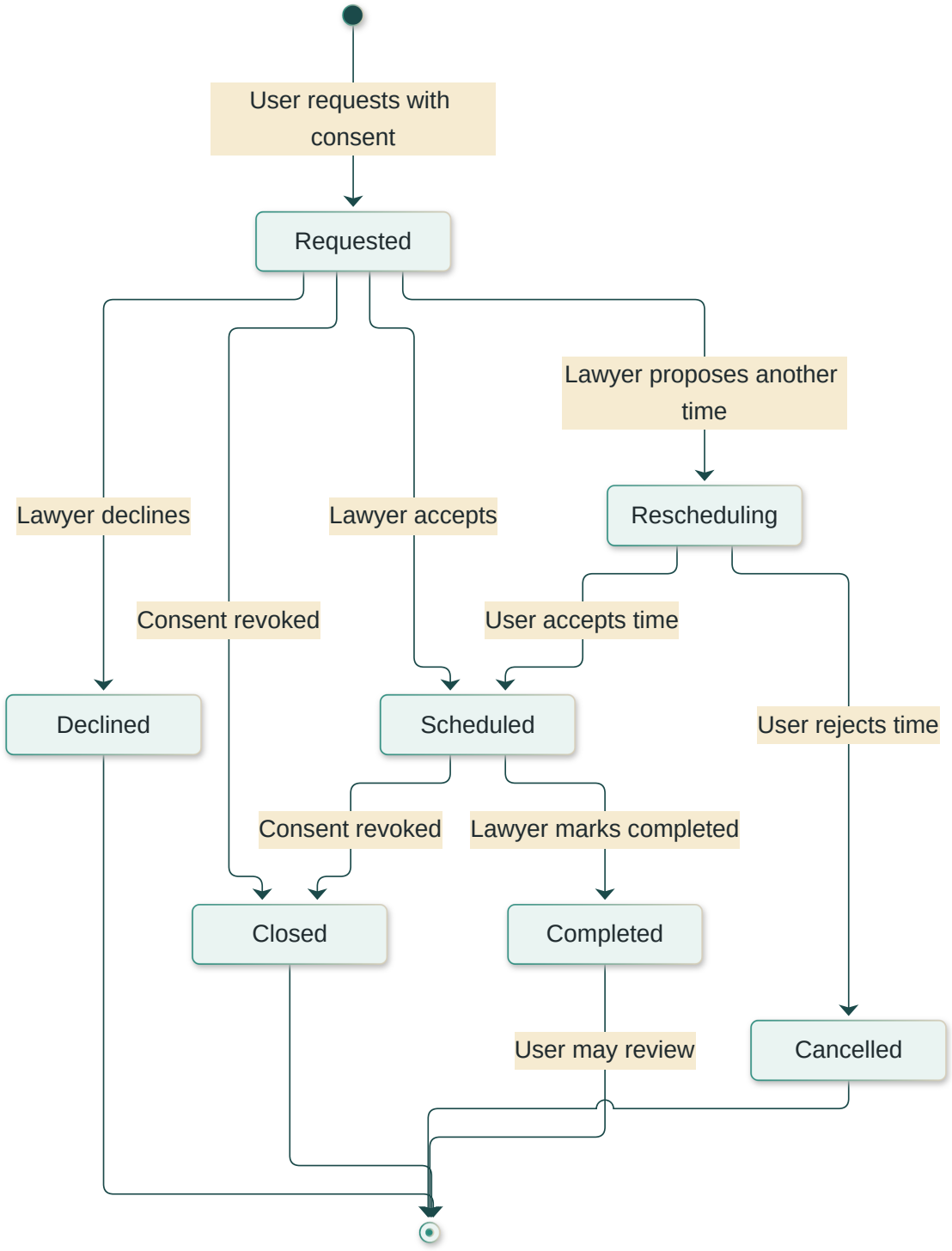


Figure 6.3: Consultation status lifecycle

6.4 Lawyer Verification Lifecycle

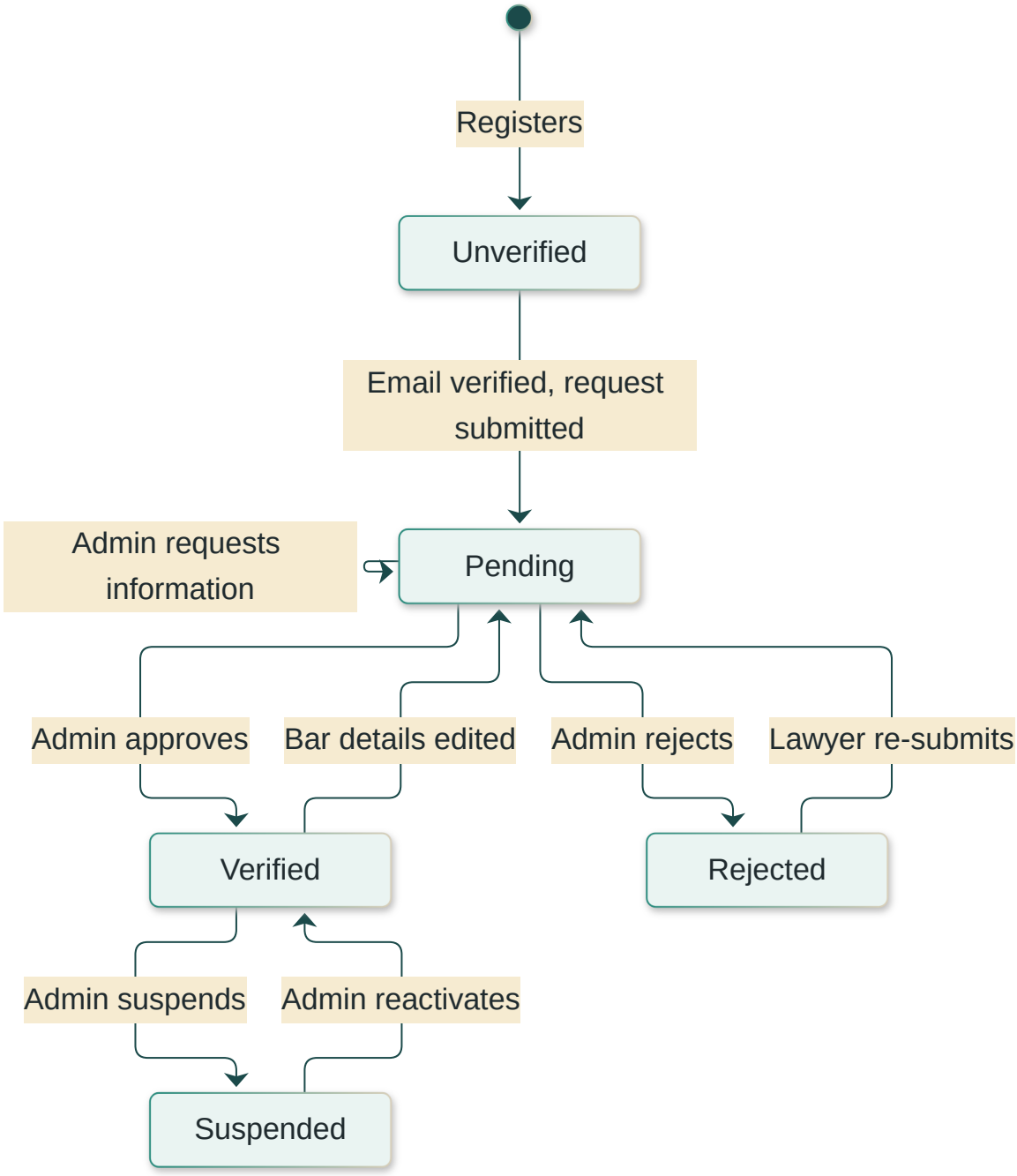


Figure 6.4: Lawyer verification lifecycle

6.5 Legal Source Lifecycle

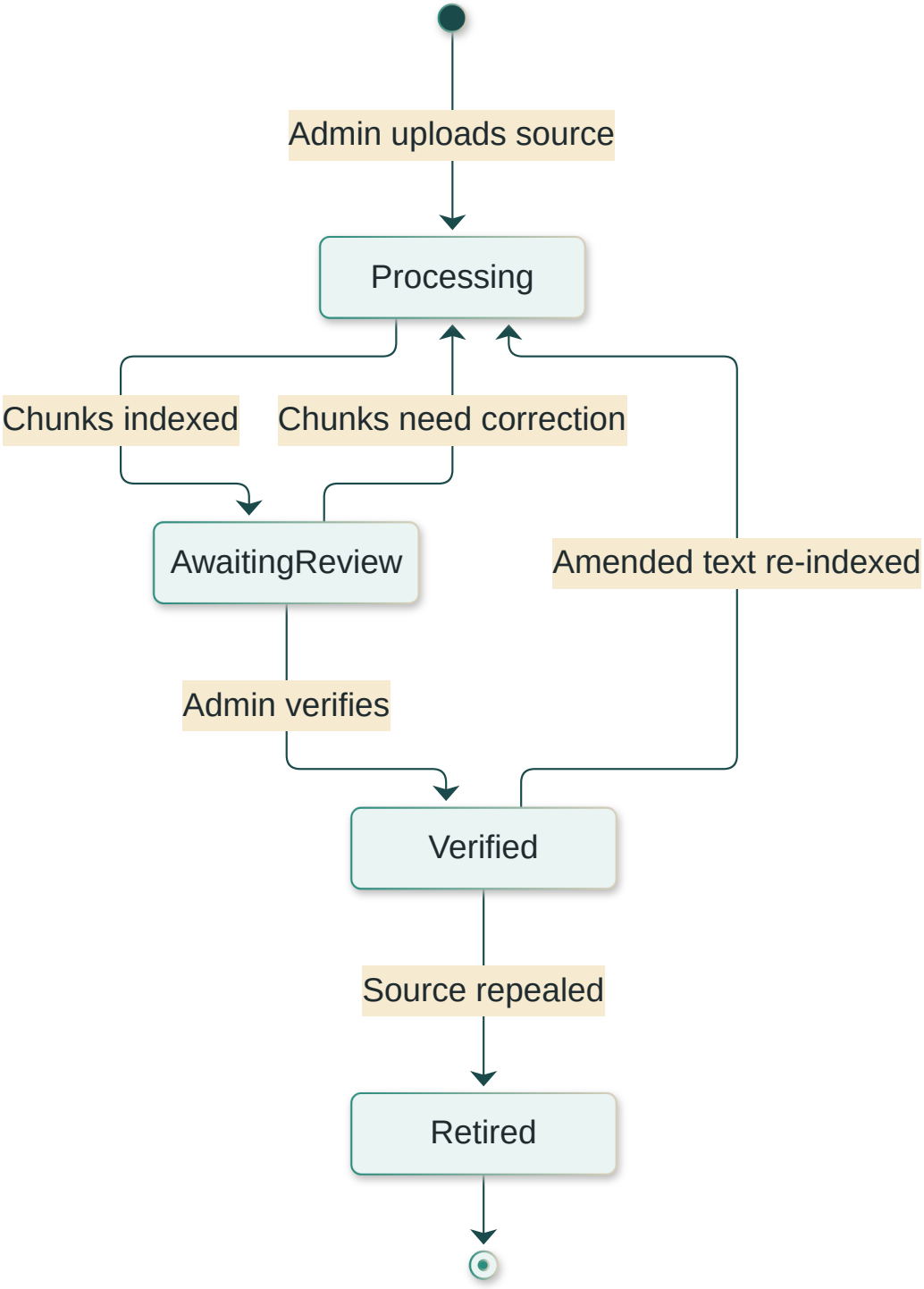


Figure 6.5: Legal source lifecycle

Only sources in the **Verified** state are retrievable by S-WF1. A **Retired** source is excluded from retrieval but kept so that earlier citations remain traceable.